IMDb movie review dataset: EDA (AI Assisted)

Introduction

This report summarizes exploratory data analysis (EDA) of the Stanford Large Movie Review Dataset ("aclImdb"), as used in the CIND860 SHAP sentiment-interpretability project. The dataset consists of IMDb movie reviews labeled as positive (`pos`) or negative (`neg`) based on their star rating, split into `train` and `test` partitions.

Dataset overview

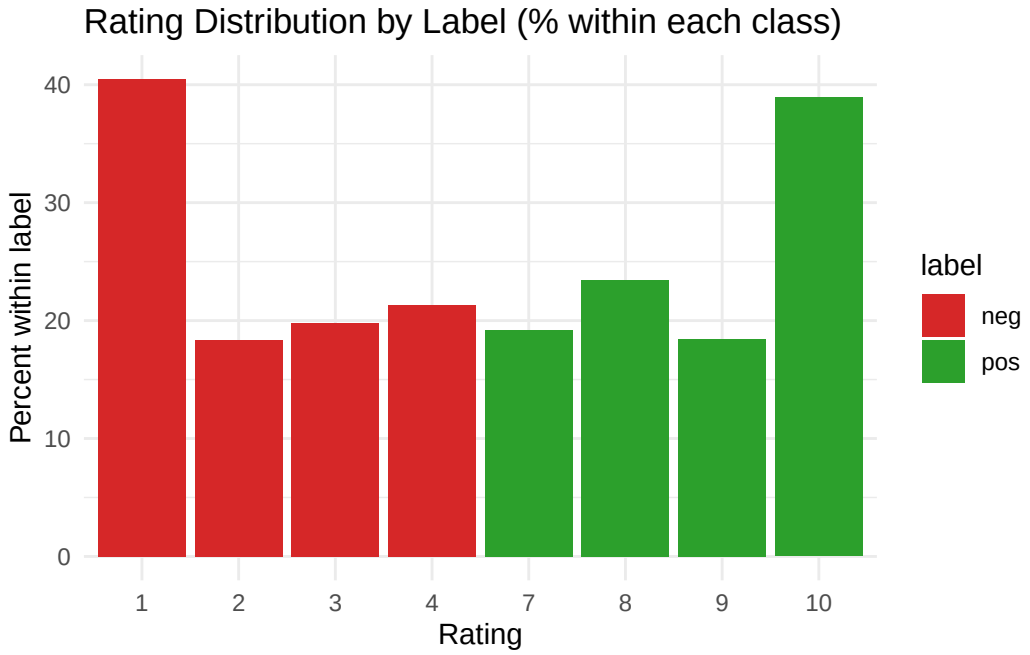
The cleaned, deduplicated dataset currently on disk contains 49877 rows and 6 columns: `text_raw`, `text_clean`, `label`, `split`, and `rating`.

```
# A tibble: 2 x 3
  split   neg   pos
  <chr> <int> <int>
1 test  12500 12500
2 train 12405 12472
```

Class balance is close to 50/50 in both splits, with a slight excess of `pos` reviews in `train` due to earlier deduplication having removed more `neg`-side duplicates.

Rating distribution by label

Review ratings range from 1–10, with ratings 5 and 6 intentionally excluded (treated as "neutral" and not included in the labeled dataset). Negative reviews are rated 1–4; positive reviews are rated 7–10.

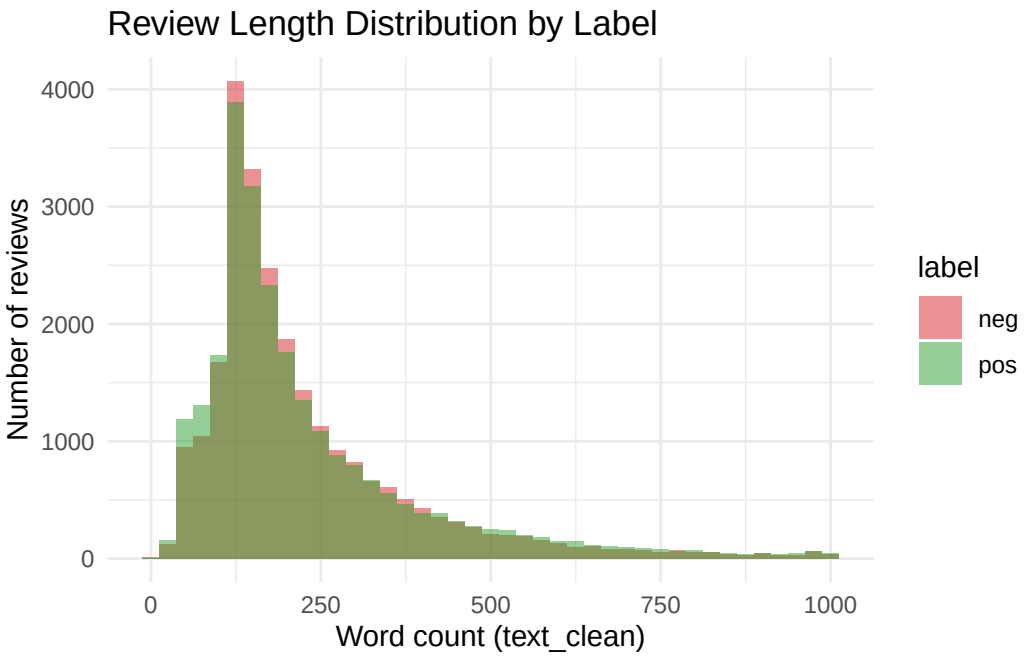


The distribution within each class is strongly U-shaped: most negative reviews are rated at the extreme low end (1), and most positive reviews are rated at the extreme high end (10), with the intermediate ratings (2–4 for negative, 7–9 for positive) making up smaller, fairly even shares.

Review length distribution by label

```
# A tibble: 2 x 5
  label  mean median   max     n
  <chr> <dbl>  <dbl> <int> <int>
1 neg    229.    174  1522 24905
```

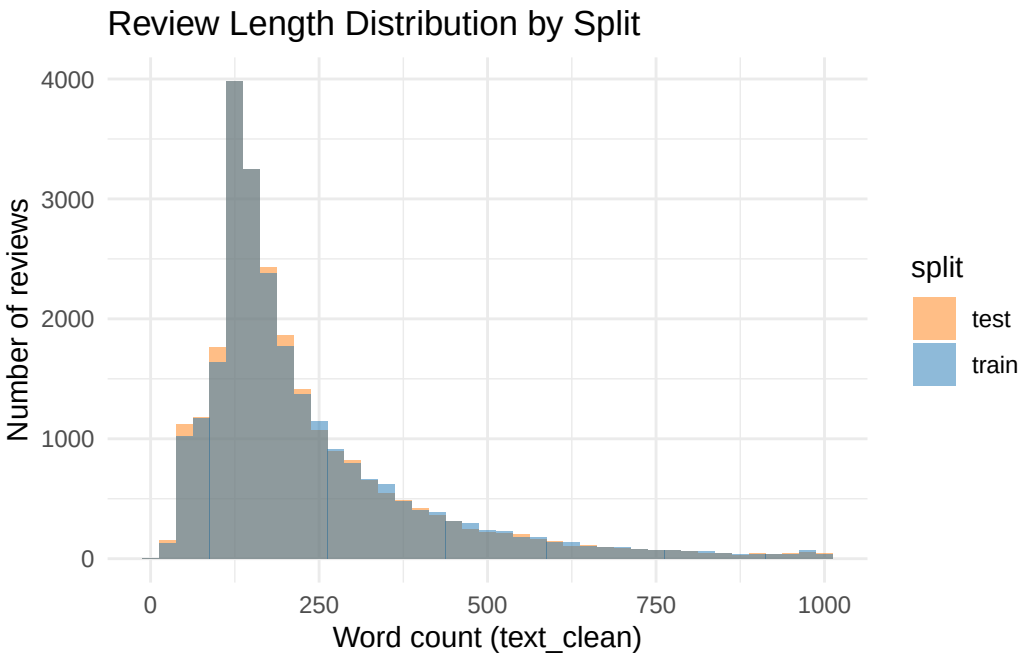
```
2 pos    233.    172  2473 24972
```



Review length, measured as word count on the cleaned text, is nearly identical between positive and negative reviews. Both distributions are heavily right-skewed, with the bulk of reviews falling between roughly 100 and 300 words and a long tail extending out to well over 1,000 words. This suggests review length alone is not a useful signal for distinguishing sentiment.

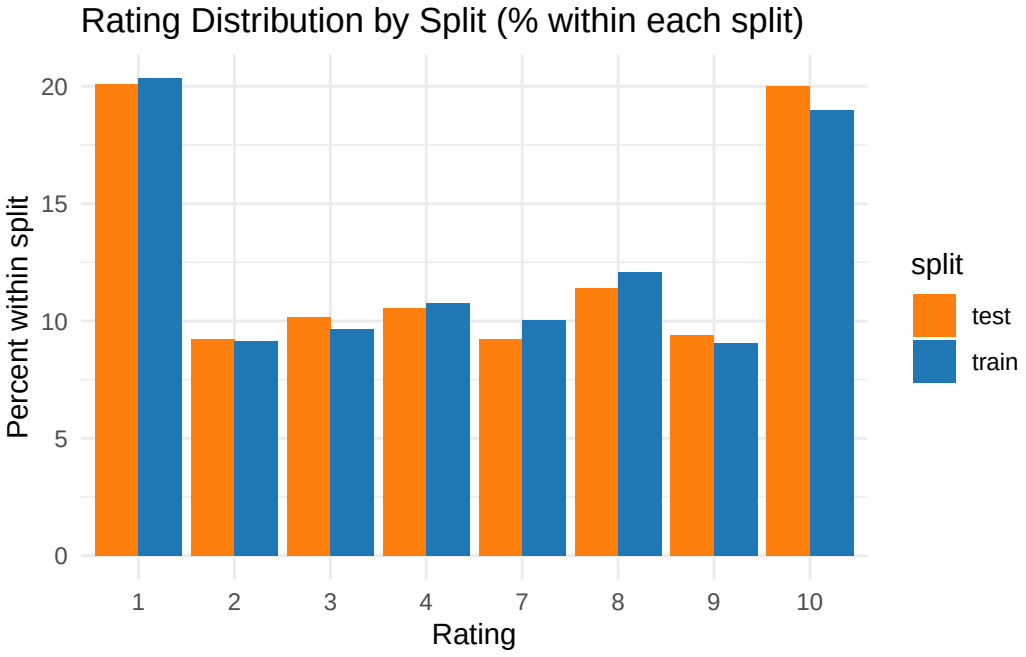
Review length distribution by split

```
# A tibble: 2 x 5
  split  mean median   max     n
  <chr> <dbl>  <dbl> <int> <int>
1 test   228.    172  2246 25000
2 train  234.    174  2473 24877
```



Review length distributions are also very similar between the `train` and `test` splits, with closely matching means, medians, and overall shape. This indicates the train/test split does not introduce a length-based bias.

Rating distribution by split



The rating distribution is likewise consistent across splits: both `train` and `test` show the same U-shape, with roughly 19–20% of reviews at each extreme (rating 1 or rating 10) and the remaining ratings spread similarly across both splits. This confirms the split is representative and was not meaningfully disrupted by the deduplication process, which only removed rows from `train`.

Summary of findings

- **Rating strongly predicts label by construction**, and within each label the rating distribution is bimodal/U-shaped, with reviews skewing toward the extremes (1 or 10).
- **Review length does not distinguish sentiment**: positive and negative reviews have nearly identical word-count distributions.
- **Train and test splits are well-matched** in both review length and rating distribution, suggesting no systematic bias was introduced by the deduplication process.